

Curriculum Vitae

SUMMARY OF QUALIFICATION

- Seven years of research experience in the field of human factors (domains of healthcare and transportation)
- Three years of teaching experience as an independent instructor
- Four years of industrial experience
- Professional in programming and data analysis
- Passionate about behavioral assessments, measurement of burnout, stress, and physiological responses using wearable technologies and machine learning models

EDUCATION AND FORMAL TRAINING

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| • Postdoctoral fellow, Houston Methodist (HM) | 2020 – 2022 |
| • PhD: Engineering Management, Western New England University, MA | 2015 – 2020 |
| • MS: Engineering Management, Western New England University, MA | 2014 – 2015 |
| • BSc.: Mechanical Engineering, Azad University Central Tehran Branch, Tehran, Iran | 2001 – 2006 |

EXPERIENCE

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| 1- Teaching Professor at Rensselaer Polytechnic Institute (RPI) | 2022 – Present |
| ➤ Duties | |
| - Teaching and developing undergraduate and graduate courses | |
| - Chief Engineer at Capstone Projects | |
| - Leading a team to attend the HSPI healthcare simulation competition | |
| 2- Faculty appointment at the department of anesthesia and Critical Care, Houston Methodist (HM) | 2022 – Present |
| - Active in ICU-related studies on intensivists | |
| 3- Postdoctoral Fellow at Houston Methodist (HM) | 2020 – 2022 |
| ➤ Research Focus | |
| ▪ Burnout and stress among cardiovascular intensive care unit (CVICU) nurses and physicians using wearable technologies. | |
| ▪ Evaluation of nurses' workload and stress using physiological and ocular data | |
| ▪ Measurement of physical activities of ICU nurses using accelerometer data | |
| ▪ Study of the Mindfulness-Based Stress Reduction (MBSR) courses on clinicians' mental health | |
| ▪ Web-based communication platform on patients' behavior (e.g., no-show) | |
| ▪ Development of a tailored application on medical compliance of African American breast cancer patients who takes hormonal therapy. | |
| 4- Research Assistant at Western New England University | 2014 – 2016 |
| ▪ Four-year research experience related to training and behavior assessment. | |
| ▪ Collaborated in the development of a tablet-based driving program—Engaged Driver Training System (EDTS)—for teen drivers. | |
| ▪ Designed a gaze-based training intervention for student pilots of general aviation. | |
| ▪ Analyzed the National Transportation Safety Board (NTSB) aircraft accident database by developing an application using Visual Basic.NET. | |

SELECTED PUBLICATIONS

Published

- 1- **Ahmadi N**, Yang J, Sasangohar, F, Danesh V, Denney Y, Masud F, Quantifying Workload and Stress in Intensive Care Unit Nurses Using Naturalistic Evaluation of Clinical Care Delivery, The Journal of the Human Factors and Ergonomics Society, 2022.
- 2- **Ahmadi N**, Sasangohar F, Nisar T, Danesh V, Larsen E, Sultana I, Bosetti R. Quantifying Occupational Stress in Intensive Care Unit Nurses: An Applied Naturalistic Study of Correlations Among Stress, Heart Rate, Electrodermal Activity, and Skin Temperature. Hum Factors. 2022 Feb;64(1):159-172. doi: 10.1177/00187208211040889
- 3- **Ahmadi N**, Romoser M, Salmon C. Improving the tactical scanning of student pilots: A gaze-based training intervention for transition from visual flight into instrument meteorological conditions. Appl Ergon. 2022 Apr;100:103642. doi: 10.1016/j.apergo.2021.103642
- 4- Dhala A, Sasangohar F, Kash B, **Ahmadi N**, Masud F. Rapid Implementation and Innovative Applications of a Virtual Intensive Care Unit During the COVID-19 Pandemic: Case Study. J Med Internet Res. 2020 Sep 3;22(9):e20143. doi: 10.2196/20143
- 5- Sasangohar F, Dhala A, Zheng F, **Ahmadi N**, Kash B, Masud F. Use of telecritical care for family visitation to ICU during the COVID-19 pandemic: an interview study and sentiment analysis. BMJ Qual Saf. 2021 Sep;30(9):715-721. doi: 10.1136/bmjqs-2020-011604

Submitted:

- 6- **Ahmadi N**, Larsen E, Rao A, Danesh V, Faisal M, Sasangohar F, A Novel Protocol for High Fidelity Observation of Stress in Intensive Care Nursing through Wearable Technology, Under Review, BMJ Quality & Safety.
- 7- **Ahmadi N**, Tariq N., Shomaker R., Sasangohar F., Sheth P., Montgomery T., Assessment of a Live Chat Communication System and Text Messaging Reminder on Outpatient Appointment Compliance In a Large Healthcare Setting: a Cross-Sectional Study, BMJ Quality & Safety.

In Progress

- 8- **Ahmadi N**, Yang J, Sasangohar, F, Danesh V, Denney Y, Masud F, Exploring the effect of a Code Blue Event on Intensive Care Unit Nurses' Stress and Physiological Responses: A Naturalistic Case Study.
- 9- **Ahmadi, N.**, Katrahmani, A., Romoser, M., Commentary Driving Assessment on a Tablet-based Training Program.

Conference Proceedings:

- 10- **Ahmadi, N.**, Katrahmani, A. Romoser, M. Short and long-term transfer of training in a tablet-based teen driver hazard perception training program. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. 2018;62:1965-1969. doi: 10.1177/1541931218621445
- 11- **Ahmadi, N.**, Romoser, M., Guarnieri, L., Kry, T., Porter-Fyke, E. Application of Eye Tracking Technology in Naturalistic Usability Assessment of an Academic Library Website. Published in: *Advances in Usability, User Experience and Assistive Technology: Proceedings of the AHFE 2018 International Conferences ..., Held on July 21–25, 2018, in Loews Sapphire Falls Resort at Universal Studios, Orlando, Florida, USA*. Springer International Publishing; 2018.
- 12- Katrahmani, A., **Ahmadi, N.** Romoser, M. Using Situation Awareness as a Measure of Driver Hazard Perception Ability, Proceeding paper at 9th International Driving Symposium on Human Factors in Driver Assessment, Training and Vehicle Design, 2017 June 28, Manchester Village, Vermont, USA. doi:10.17077/drivingassessment.1644

Curriculum Vitae - 2026

- 13- Demirtas, M., **Ahmadi, N.**, Dehghanimohammadabadi, M., Highlighting the Main Factors of Internet Banking via Multiple Criteria Decision Analysis, Proceedings of the 2016 Industrial and Systems Engineering Research Conference (ISERC), **Best track paper of Engineering Management Society.**

CONFERENCE PRESENTATIONS

- 1- **Ahmadi, N.**, Sasangohar, F., Taming no-show rate; exploring the impact of facilitating communication with patients using a live chat system and text reminders on patients' appointment compliance, Human Factors and Ergonomics Society (HFES) 2022.
- 2- **Ahmadi, N.**, Yang J, Sasangohar, F, Danesh V, Denney Y, Masud F, Application of eye movements metrics in naturalistic assessment of intensive care unit nurses' mental workload, Human Factors and Ergonomics Society (HFES) 2022.
- 3- McNulty, D., Sasangohar, F., Jones-Wood, S., **Ahmadi, N.**, Evaluating the effectiveness of virtual mindfulness-based stress reduction, Healthcare Systems Process Improvement (HSPI) 2021, Virtual.
- 4- **Ahmadi, N.**, Pooja, S., LeTessa, M., Sasangohar, F., The impact of a web-based scheduling and communication system on the rate of broken appointments, Healthcare Systems Process Improvement (HSPI) 2021, Virtual.
- 5- **Ahmadi, N.**, Sasangohar, F., Insight into the application of Tele-Critical Care for family visitation in the COVID-19 pandemic, Healthcare Systems Process Improvement (HSPI) 2021, Virtual.
- 6- Jahroumi, R., **Ahmadi, N.**, Sasangohar, F., Danesh, V., Workload intensity in the intensive care unit: naturalistic physical activities assessment using accelerometer data, Healthcare Systems Process Improvement (HSPI) 2021, Virtual.
- 7- **Ahmadi, N.**, Improving the tactical scanning of student pilots, TRB Annual Meeting, 2020 ACRP AVO90(1) Sub-committee for Aviation Safety, Immerring Research January 12-16, Washington, DC

WEBINAR

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| 1- Tobii Pro webinar on the application of eye-tracking technology in various domains | December 2021 |
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WORKSHOP

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| 1- Summer Machine learning workshop EnMed students | July 2020 |
| 2- Summer Machine learning workshop EnMed students | July 2021 |

Grant

A \$10,000 Begi grant was awarded to study the impact of COVID-19 on ICU nurses via collecting physiological data in a naturalistic study and interviewing nurses.

TEACHING EXPERIENCES

Independent Instructor:

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| 1- ISYE 4260: Human Performance Modeling
(Instructor, In-class, Undergraduate, one time)
Rensselaer Polytechnic Institute
Department of Industrial and Systems Engineering | Fall 2022 |
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Syllabus: models of information processing and situation awareness, mental workload assessments, modeling user behavior (e.g., GOMS, KLM, NGOMSL) using CogTool and Cogulator, application of wristband technology (Empatica E4), eye tracking device (Tobii Pro Glasses 2), HTC VIVE Pro Virtual Reality, and a flight simulator to evaluate human performance; applied statistic needed for analysis of human performance in Python.

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| 2- ISYE 4210/ ISYE 6600: Design and Analysis of Supply Chains
(Instructor, In-class, Undergraduate and graduate level, one time) | Fall 2022 |
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Rensselaer Polytechnic Institute
Department of Industrial and Systems Engineering

Syllabus: the design and operation of supply chain operations, demand forecasting, aggregate planning, coordination, inventory management, routing, and scheduling in transportation.

- 3- ISYE-4290/6620: Discrete Event Simulation Spring 2022
(Instructor, In-class, Undergraduate and graduate level, two times)
Rensselaer Polytechnic Institute
Department of Industrial and Systems Engineering

Syllabus: Random number generators, random variates, Monte Carlo Simulation, simulation in Excel, manufacturing and healthcare simulation in FlexSim, verification, and validation, input and output Analysis.

- 4- ISYE-4330/6020: Design of Experiments Spring 2022
(Instructor, In-class, Undergraduate and graduate level, two times)
Rensselaer Polytechnic Institute
Department of Industrial and Systems Engineering

Syllabus: One- and two-way ANOVA, statistical test (Tukey, Fisher, and ...), general factorial design, 2^k factorial design, replicated, unreplicated, and duplicated factorial design, fractional factorial design, and power and sample size analysis.

- 5- ISYE 4270/ECSE 4900/MTLE 4920: Multidisciplinary Capstone Design Spring 2022
(Chief Engineer, In-class, Undergraduate level, one time)
Rensselaer Polytechnic Institute
Department of Industrial and Systems Engineering

- 6- IE 212: Probability & Statistics 2016 – 2019
(Instructor, In-class, Undergraduate level, six times)
Western New England University
Department of Industrial Engineering and Engineering Management

Syllabus: Learning concepts of probability, different probability distributions and their applications, basic statistical analysis, tests of hypothesis, and pair-wise comparison tests.

- 7- IE 419 Industrial Engineering Computer Applications 2018 – 2019
(Instructor, In-class, Undergraduate level, two times)
Western New England University
Department of Industrial Engineering and Engineering Management

Syllabus: Programming and simulating different processes, basic statistical analysis, and tests of hypothesis using VBA, Python, and SQL.

- 8- ME 205-41 Measurement Computing 2017 – 2018
(Instructor, In-class, Undergraduate level, three times)
Western New England University
Department of Mechanical Engineering

Curriculum Vitae - 2026

Syllabus: Programming in Visual Basic.NET and Visual Basic Application, designing forms, database coding, performing engineering calculations, and visualization of results.

Teaching Assistant

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|---|-----------|
| 1- Probability & Statistics
Western New England University
Department of Industrial Engineering and Engineering Management | Fall 2016 |
| 2- Facility Design & Material Handling
Western New England University
Department of Industrial Engineering and Engineering Management | Fall 2015 |

Guest Lecturer

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|---|-------------|
| 1- Human Factors Design Engineering
Western New England University
Department of Industrial Engineering and Engineering Management | Fall 2016 |
| 2- Work Analysis and Design
Western New England University
Department of Industrial Engineering and Engineering Management | Fall 2016 |
| 3- Design of Experiment
Western New England University
Department of Industrial Engineering and Engineering Management | Spring 2016 |
| 4- Facility Design and Material Handling
Western New England University
Department of Industrial Engineering and Engineering Management | Fall 2015 |

ACADEMIC RESEARCH EXPERIENCE AND PROJECTS

- Barriers to commercializing renewable energy technologies in Africa, Asia, Europe, and North America
- Water treatment plant cost estimation

INDUSTRY EXPERIENCES

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| 1- Project Manager and Coordinator of Piping Dep., APEC, Tehran, Iran
▪ Experienced various aspects of basic and detailed engineering design
▪ Customized SharePoint for Project Management | June 2010 – June 2014 |
| 2- Manager of Heat Exchanger Dep., COOLSUMMER, Tehran, Iran
▪ Led design and repair of Plate Heat Exchanger (PHE) | January 2009 – June 2010 |

SOFTWARE SKILLS

- **Coding:** Python, Visual Basic.NET, Visual Basic for Applications (VBA)
- **Database:** SQL Server, Access
- **Data analysis:** Python, R, Minitab, Excel
- **Simulation:** FlexSim (Healthcare and Manufacturing)
- **Project Management:** MS Project
- **Optimization:** Excel Solver
- **Plant Design:** AVEVA PDMS

Curriculum Vitae - 2026

- **Plant review:** Navisworks Simulate, Navisworks Management
- **2D Drawing:** AutoCAD
- **General Software:** MS Office

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

- Human Factors and Ergonomics Society (HFES)

SELECTED COURSEWORK

- Courses during Ph.D. degree: Advanced Cost Estimation, Factory Design & Material Handling, Discrete Event Simulation, Quality System Process Improvement, Engineering Psychology, Seminar / Research Methods for Engineering Management
- Courses during the Master's degree: Design of Experiment, Multi-Criteria Decision Analysis, Optimization Method I, Quality Engineering, Engineering Cost Analysis, Statistical Quality Control, Engineering Risk Management, Engineering Supply Chain

HONORS AND AWARDS

- Best Paper Student Competition Award, 3rd Place 2016
"Highlighting the Main Factors of Internet Banking via Multiple Criteria Decision Analysis Presented to Engineering Management Society for Engineering and Management Systems (SEMS)
- Engineering Research Assistance Scholarship, Western New England University 2016

CERTIFICATES

- Six Sigma Green Belt Program, Western New England University 2017

DATA ANALYSIS CERTIFICATES

- Springboard: Intermediate Data Science: Python 2018 – 2019
Syllabus: data wrangling, data storytelling, explanatory data analysis, data visualization, data analysis, supervised and unsupervised machine learning.
- Machine Learning with Python 2020
Syllabus: machine learning models including support vector machine, decision tree, and binary logistic regression.
- Neural Networks and Deep Learning (In progress) 2020
Syllabus: the foundations of deep learning and implementation of efficient (vectorized) neural networks.